

HUNTER KRAUKLIS

Position: Undergraduate/Post-Baccalaureate Researcher, University of Wisconsin–Madison

Madison, WI

Research Interests: Exoplanet Dynamics, Galactic Dynamics & Dark Matter, Philosophy of Physics

hkrauklis@wisc.edu

Methods: Observational & Computational Astrophysics, Experimental Instrumentation, Embedded Systems/Control

Statement

I graduated from the University of Wisconsin–Madison in May 2026 with a B.A. in Astronomy–Physics and Philosophy, and I continue at my alma mater as an undergraduate/post-baccalaureate researcher. I work with Professor Juliette Becker on exoplanet dynamics and planetary system formation, and with Professor Victor Brar on experimental instrumentation for condensed-matter and precision-metrology applications — including a quasi-zero-stiffness flexure system for isolating a cryogenic scanning tunneling microscope from vibration. I also tutored introductory physics students at the Physics Learning Center. Alongside my physics and astronomy training, I bring a background in the philosophy of science, quantum mechanics, and metaphysics, which shapes how I approach open conceptual questions in astrophysics.

Research Experience

Undergraduate/Post-Baccalaureate Researcher

January 2026 – Present

Prof. Juliette Becker, Department of Astronomy, University of Wisconsin–Madison

- Conduct research in exoplanet dynamics and planetary system formation, supporting studies of orbital architectures and long-term system stability
 - Analyze astrophysical data on exoplanet orbits, protoplanetary disk interactions, and planetary climate
 - Assist with computational modeling and dynamical analysis of planetary systems to investigate formation pathways, orbital evolution, and habitability conditions
 - Extended this work through an independent directed study (ASTRON 699, Spring 2026)
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Undergraduate/Post-Baccalaureate Research Assistant

August 2025 – Present

Prof. Victor Brar, Department of Physics, University of Wisconsin–Madison

- Fabricate high-quality graphene and hexagonal boron nitride (hBN) samples via mechanical exfoliation (“tape peeling”) for condensed-matter and nanomaterials research
 - Assist in the design and construction of a custom airtight purge enclosure integrating an FTIR spectroscopy system with microscopy equipment for measurements under controlled environmental conditions
 - Collaborated with two fellow lab members to design, build, and program an Arduino-controlled, position-encoding xyz stage — including a custom breadboard circuit, linear actuator, and Python/C++ control and real-time position-tracking software — replacing a manual, wobble-prone setup and enabling precise, repeatable displacement testing of quasi-zero-stiffness (QZS) flexures for a cryogenic scanning tunneling microscope vibration-isolation system
 - Support experimental setup development, sample preparation, and laboratory instrumentation for nanoscale optical and materials-characterization studies
 - Contribute to science-outreach initiatives focused on accessibility for blind and visually impaired audiences
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Outreach & Research Volunteer

September 2021 – August 2022

Geneva Lakes Astrophysics and STEAM (GLAS), Walworth, WI

- Developed foundational proficiency in Python for observational and data-related astronomy applications
- Trained to remotely operate the Stone Edge Observatory telescope for research and educational purposes
- Supported telescope use for astronomical observation and data collection activities
- Delivered public education and outreach on a range of astrophysics topics, communicating complex scientific concepts to diverse audiences

Presentations

“Dancing with the Stars: Measuring the Rotation Curve of the Milky Way Galaxy”

April 2026

Poster — 2026 UW–Madison Undergraduate Symposium, Madison, WI

Co-authors: A. Alvin, N. Broderick, I. Bubnick, H. Krauklis, P. Kyriopoulos, A. Nelson, N. Talbi

- Used 21-cm HI-line observations from the UW Small Radio Telescope (Galactic longitudes 22–72°) with the tangent-point method to construct the Milky Way’s rotation curve and enclosed-mass profile
- Measured a rotation-curve plateau consistent with published results, providing evidence for dark matter and disagreement with the mass distribution predicted by luminous matter alone

Outreach & Public Engagement

WSBVI College & Career Readiness Resource Fair — Invited Presenter

May 2026

With Prof. Victor Brar and two fellow Brar Lab members — Wisconsin School for the Blind & Visually Impaired, Janesville, WI

- Co-presented an interactive physics outreach session for blind and low-vision middle and high school students at WSBVI’s third annual College and Career Readiness Resource Fair
- Framed physics as a uniquely accessible field of study while leading hands-on tactile demonstrations of atomic structure and interatomic forces
- Drew on the lab’s NSF-funded “Haptic Atomic Forces” curriculum (tactile 3D-printed and crocheted electron-cloud models, raised haptic images, and a force-feedback haptic pen) to let students physically experience concepts in atomic physics

Education

B.A. in Astronomy–Physics and Philosophy

2022 – 2026

University of Wisconsin–Madison — Degree conferred May 2026

- Major GPA (Astronomy–Physics): 3.13/4.0

Honors & Awards

Dean’s List, University of Wisconsin–Madison

Spring 2026

Teaching & Mentoring Experience

Peer Mentor Tutor

August 2025 – May 2026

Physics Learning Center, University of Wisconsin–Madison

- Facilitate weekly peer-led tutoring sessions for 5–8 Physics 104 students, reinforcing core introductory physics concepts and problem-solving techniques
- Guide students through active-learning exercises, collaborative problem sets, and exam preparation strategies
- Explain complex physics topics clearly and adaptively, promoting peer-to-peer learning and academic confidence
- Foster an inclusive, supportive learning environment encouraging participation across diverse academic backgrounds

Skills & Courses

Relevant Coursework — Physics & Astronomy

ASTRON 540 Exoplanets	Spring 2026
ASTRON 699 Directed Study (Exoplanet Dynamics)	Spring 2026
ASTRON 465 Observational Astronomy	Fall 2025
PHYSICS 531 Intro to Quantum Mechanics	Fall 2025
PHYSICS 415 Thermal Physics	Spring 2026
PHYSICS 322 Electromagnetic Fields	Spring 2025
PHYSICS 311 Mechanics	Fall 2024
ASTRON 335 Cosmology	Fall 2024
ASTRON 320 The Interstellar Medium	Spring 2024

Relevant Coursework — Philosophy of Science & Logic

PHILOS 562 Philosophy of Quantum Mechanics (Special Topics in Metaphysics)	Spring 2025
PHILOS 520 Philosophy of Natural Sciences	Fall 2024
PHILOS 560 Metaphysics	Spring 2026
PHILOS 211 Elementary Logic	Fall 2023

Programming & Computation

Python · C++ (Arduino/embedded control systems) · Wolfram Mathematica · REBOUND N-body simulations · IllustrisTNG50 simulation data

Observational Techniques

Optical and radio telescope operation and data acquisition — UW Small Radio Telescope (21-cm HI spectroscopy), Stone Edge Observatory, ASTRON 465 Observational Astronomy Lab

Laboratory & Instrumentation

Mechanical exfoliation (graphene/hBN sample fabrication) · FTIR spectroscopy instrumentation · custom circuit design (breadboard, Arduino, actuators) and position-encoding control systems · experimental setup design and materials characterization

Other

Microsoft Office Suite

Github